DEPARTMENT OF AEROSPACE ENGINEERING, SMME, NUST, ISLAMABAD

Project Proposal

CS – 117

Applications of ICT

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Automated Student Identification System Using Python-Based Face Detection and Barcode Scanning

1. Introduction

This project proposes an automated system to identify students using Python-based face detection and barcode scanning. It aims to reduce congestion, increase verification speed, and minimize human error at university gates.

2. Problem Statement

The current manual verification process leads to long queues, errors, and slower gate entry. Automation is required for efficiency and accuracy.

The daily traffic congestion at the NUST entry gates has become a consistent bottleneck for students and security staff. Every morning, long queues form because the verification process relies heavily on manual checking, where guards visually confirm student identity or manually inspect student cards. This method not only slows down the entire movement of students but also increases human error and unnecessary delays. The problem becomes more severe during peak hours when hundreds of students enter at the same time, and the entire system depends on the attention, speed, and accuracy of the guards. Our project aims to target this exact issue by proposing a simple but powerful automation system built entirely in Python.

We want to design a setup that recognizes students through facial detection, and in cases where facial recognition fails, the system will allow alternative identification through scanning the barcode printed on the student card. This dual approach ensures that the entry workflow becomes faster, smoother, and significantly more reliable compared to the existing manual method.

3. Project Objectives

The purpose of this project is to give a functional demonstration of how such a system can operate using Python and commonly available libraries without depending on any complex hardware. The approach is intentionally kept minimalistic so that the entire concept stays within the scope of the course while still addressing a real-world engineering problem.

- Automate student identification using Python.
- Use face detection as the primary verification method.
- Use barcode scanning as a fallback option.
- Reduce congestion and human dependency
- Demonstrate a functional prototype.

4. System Overview

Module	Description
Face Detection	Detects and verifies student faces using OpenCV and Python.
Barcode Scanning	Fallback system using barcode/QR scanning.
Integration Layer	Handles workflow, switching, and result display.

The working principle of the final system is simple enough for demonstration yet strong enough to show the core concept: reducing the traffic congestion at NUST gates by automating identity verification.

To keep the prototype manageable, the face dataset will be limited and manually curated, while the barcode scanning portion will rely on decoding printed barcodes using Python libraries.

5. Expected Outcomes

By proposing a Python-based system that automates student verification through face detection and barcode fallback, we attempt to solve a practical problem that the university faces every single day. The project remains within course limitations, avoids complex hardware, and emphasizes software logic, clarity, and practical applicability. Even in prototype form, the idea highlights how a simple computational approach can significantly improve a real institutional process and reduce inefficiencies in daily workflow.

- Working prototype with real-time face detection.
- Reliable barcode fallback system.
- Reduced verification time.
- Full documentation and workflow explanation.

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